UNIVERSITY OF KELANIYA
DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR 2022/2023
BACHELOR OF SCIENCE (GENERAL) DEGREE
PMAT 21263 - LINEAR ALGEBRA

Semester I | Tutorial 09 | Due date : 02.08.2024 at 4.00 p.m.

Submit only 02) a, d, j, k 03), 05), 06)a, 07)

1. Let A be a given $m \times n$ matrix. Show that the function $f : \mathbb{R}^n \to \mathbb{R}^m$ defined by $f(x) = Ax$, for all $x \in \mathbb{R}^n$, is a linear transformation.

2. Determine which of the following functions are linear transformations. Prove that your answers are correct. Which are linear operators?

   (a) $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by $f([x, y]) = [3x - 4y, -x + 2y]$
   (b) $h : \mathbb{R}^4 \to \mathbb{R}^4$ given by $h([x_1, x_2, x_3, x_4]) = [x_1 + 2, x_2 - 1, x_3, 3]$
   (c) $k : \mathbb{R}^3 \to \mathbb{R}^3$ given by $k([x_1, x_2, x_3]) = [x_2, x_3, x_1]$
   (d) $l : M_{2\times2} \to M_{2\times2}$ given by $l\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{pmatrix} a - 2c + d & 3b - c \\ -4a & b + c - 3d \end{pmatrix}$
   (e) $n : M_{2\times2} \to \mathbb{R}$ given by $n\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$
   (f) $r : P_3 \to P_2$ given by $r(ax^3 + bx^2 + cx + d) = (a)^{\frac{1}{3}}x^2 - b^2x + c$
   (g) $s : \mathbb{R}^3 \to \mathbb{R}^3$ given by $s([x_1, x_2, x_3]) = [\cos x_1, \sin x_2, e^{x_3}]$
   (h) $t : P_3 \to \mathbb{R}$ given by $t(a_3x^3 + a_2x^2 + a_1x + a_0) = a_3 + a_2 + a_1 + a_0$
   (i) $u : \mathbb{R}^4 \to \mathbb{R}$ given by $u([x_1, x_2, x_3, x_4]) = |x_2|$
   (j) $v : P_2 \to \mathbb{R}$ given by $v(ax^2 + bx + c) = abc$
   (k) $g : M_{3\times2} \to P_4$ given by $g\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} = a_{11}x^4 - a_{21}x^2 + a_{31}$
   (l) $e : \mathbb{R}^2 \to \mathbb{R}$ given by $e([x, y]) = \sqrt{x^2 + y^2}$

3. Consider the function $f : M_{nn} \to M_{nn}$ given by $f(A) = BA$, where B is some fixed $n \times n$ matrix. Show that f is a linear operator.

4. Let B be a fixed nonsingular matrix in M_{nn}. Show that the mapping $f : M_{nn} \to M_{nn}$ given by $f(A) = B^{-1}AB$ is a linear operator.

5. Find the kernel of each of the following transformations.

(a) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z) = (x, 0, z)$

(b) $T : \mathbb{R}^4 \to \mathbb{R}^4$, $T(x, y, z, w) = (y, x, w, z)$

(c) $T : P_2 \to \mathbb{R}$, $T(a_0 + a_1 x + a_2 x^2) = a_0$

(d) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + 2y, y - x)$

6. Let $L : \mathbb{R}^3 \to \mathbb{R}^3$ be given by

$$L\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 5 & 1 & -1 \\ -3 & 0 & 1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

(a) Is $\begin{bmatrix} 1 & -2 & 3 \end{bmatrix}$ in $\ker(L)$? Why or why not?

(b) Is $\begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$ in $\ker(L)$? Why or why not?

7. Consider the vector space $V_4^{\text{col}}(\mathbb{R}) = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$.

Let $T : V_4^{\text{col}}(\mathbb{R}) \to V_4^{\text{col}}(\mathbb{R})$ be a linear transformation defined by

$$T \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2 & -1 & 3 & -1 \\ 1 & 1 & -2 & 1 \\ 7 & 1 & 0 & 1 \\ 3 & -3 & 8 & -3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}.$$

(a) Find a basis for $\ker(T)$ and then find $\dim(\ker(T))$.

(b) Extend the basis found in part (a) to a basis of $V_4^{\text{col}}(\mathbb{R})$.

(c) Find a basis for $\text{Im}(T)$ using the extended basis from part (b).

(d) Find $\dim(\text{Im}(T))$.

(e) Justify the equation $\dim(V_4^{\text{col}}(\mathbb{R}))) = \dim(\ker(T)) + \dim(\text{Im}(T))$.